ModelIt!

Student Activity Pack

Can You Send a Message with Light?

The Science of Light Communication

Grade 1 | NGSS 1-PS4-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Draw a picture of someone sending a message to a friend far away using light.**

2. **Circle the things that use light to send a message: a traffic light, a pillow, a lighthouse, a lamp, a shoe.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Signal Brightness

Message Received

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Bright Flashlight

Signal Brightness set to high

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Dim Flashlight

Signal Brightness set to low

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can we use light to send a message to someone far away? Use evidence from your simulation to support your answers.

8 STEM Challenge: Build a Light Communicator

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YOUR ENGINEERING MISSION

Design a device that uses light to send a simple message to a partner across the room.

THE SCENARIO

Two friends live on different sides of a valley. They cannot shout loud enough to hear each other. Can you help them design a light communicator so they can send simple messages back and forth? Make a code so they know what the light flashes mean!

Engineering Design Thinking

1. WHAT WILL YOU USE AS YOUR LIGHT SOURCE?
2. HOW WILL YOU CREATE DIFFERENT SIGNALS FOR DIFFERENT MESSAGES?
3. HOW WILL YOU MAKE SURE YOUR PARTNER CAN SEE THE LIGHT FROM FAR AWAY?
4. WHAT CODE WILL YOU USE — HOW MANY FLASHES FOR EACH MESSAGE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Telecommunications Engineers design systems that send messages using light, radio waves, and signals. Even internet cables use light! They earn $55,000-$100,000/year.